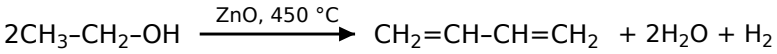
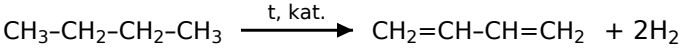
4. Olinishi

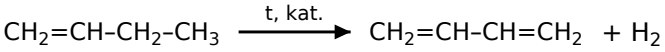
1) Lebedev usuli – etil spirtdan (MgO, ZnO yoki Al_2O_3, 400–500 °C)

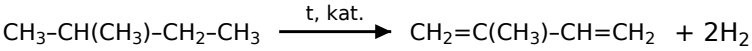


S.V. Lebedev 1928-yilda taklif qilgan; 1932-yildan butadiyen sanoatda shu usulda olina boshlagan.

2) Alkanlar va alkenlarni degidrogenlash (Cr_2O_3/Al_2O_3, ≈ 600 °C)

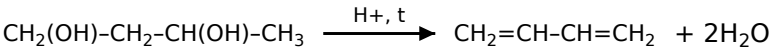




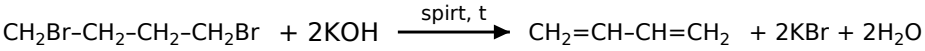


(izopentan → izopren)

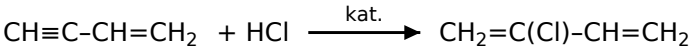
3) Glikollarni degidratlash



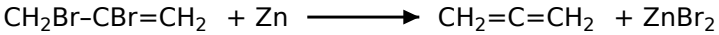
4) Digalogenalkanlarga KOH ning spirtli eritmasi ta'siri



5) Xloropren – vinilatsetilenga HCl biriktirib



6) Allen – 2,3-dibrompropenga rux ta'siri



5. Kimyoviy xossalari

Ikkita π-bog' borligi uchun diyenlar alkenlarga o'xshab **birikish**, **oksidlanish** va **polimerlanish** reaksiyalariga kirishadi. Konyugirlangan diyenlar uchun **1,2- va 1,4-birikish** xos: mahsulot diyenning tuzilishi, reaksiya sharoiti, erituvchi va reagentga bog'liq.

1,2-birikish – reagent bitta qo'sh bog'ga (C_1 va C_2 ga) birikadi. **1,4-birikish** – reagent chetki C_1 va C_4 ga birikadi, qo'sh bog' o'rtaga (C_2=C_3) ko'chadi. Past haroratda 1,2-, yuqoriroq haroratda asosan 1,4-mahsulot hosil bo'ladi.

I. Birikish reaksiyalari

1) Gidrogenlash (Ni, Pt, t)

$$CH_2{=}CH{-}CH{=}CH_2 + H_2 \xrightarrow{1,2} CH_3{-}CH_2{-}CH{=}CH_2 \quad \text{buten-1}$$

$$CH_2{=}CH{-}CH{=}CH_2 + H_2 \xrightarrow{1,4} CH_3{-}CH{=}CH{-}CH_3 \quad \text{buten-2}$$

$$CH_2{=}CH{-}CH{=}CH_2 + 2H_2 \xrightarrow{Ni,\ t} CH_3{-}CH_2{-}CH_2{-}CH_3 \quad \text{butan}$$

2) Galogenlash (I. Tile o'rgangan)

$$CH_2{=}CH{-}CH{=}CH_2 + Br_2 \xrightarrow{1,2} CH_2Br{-}CHBr{-}CH{=}CH_2$$

3,4-dibrombuten-1

$$CH_2{=}CH{-}CH{=}CH_2 + Br_2 \xrightarrow{1,4} CH_2Br{-}CH{=}CH{-}CH_2Br$$

1,4-dibrombuten-2

$$CH_2{=}CH{-}CH{=}CH_2 + 2Br_2 \longrightarrow CH_2Br{-}CHBr{-}CHBr{-}CH_2Br$$

brom mo'l bo'lsa: 1,2,3,4-tetrabrombutan. Bromli suv rangsizlanadi – sifat reaksiya.

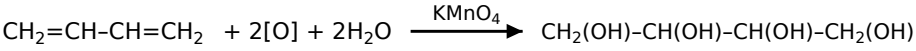
3) Galogenvodorodlarning birikishi

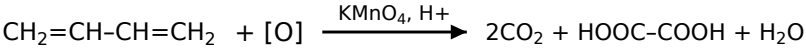
$$CH_2{=}CH{-}CH{=}CH_2 + HCl \xrightarrow{1,2} CH_3{-}CHCl{-}CH{=}CH_2 \quad \text{3-xlorbuten-1}$$

$$CH_2{=}CH{-}CH{=}CH_2 + HCl \xrightarrow{1,4} CH_3{-}CH{=}CH{-}CH_2Cl \quad \text{1-xlorbuten-2}$$

II. Oksidlanish

$KMnO_4$ eritmasini **rangsizlantiradi** (sifat reaksiya). Yumshoq oksidlanganda (Vagner) to'rt atomli spirt, qattiq oksidlanganda qo'sh bog'lar uzilib kislotalar va CO_2 hosil bo'ladi:

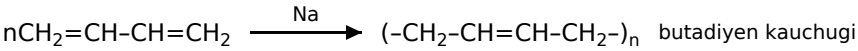


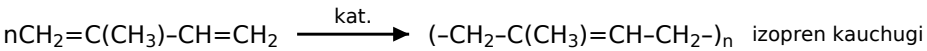


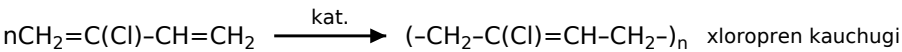
Yonishi: $2C_4H_6 + 11O_2 \rightarrow 8CO_2 + 6H_2O$

III. Polimerlanish va sopolimerlanish

Diyenlar katalizator (masalan, Na) yoki ultrabinafsha nur ta'sirida asosan **1,4-holat** bo'yicha, radikal mexanizmda oson polimerlanadi; har bir zvenoda bitta qo'sh bog' qoladi. Zvenolar sis- yoki trans-konfiguratsiyada bo'lishi mumkin.







Sopolimerlanish – ikki xil monomerning birgalikda polimerlanishi. Butadiyen + stirol (C_6H_5–CH=CH_2) → **butadiyen-stirol kauchugi**; butadiyen + akrilonitril (CH_2=CH–CN) → **butadiyen-nitril kauchugi** (moy va benzinga chidamli).

6. Kauchuklar va rezina

Tabiiy kauchuk

- Kauchukli o'simliklar (masalan, Braziliya **geveyasi**) sutsimon shirasi – **lateks**dan olinadi.
- Tarkibi **$(C_5H_8)_n$** – izopren polimeri; tuzilishi **sis-1,4-poliizopren**. Trans-1,4-poliizopren – **guttapercha** (elastik emas).
- Elastik, suv va gaz o'tkazmaydi, elektr tokini o'tkazmaydi; benzinda bo'kadi va eriydi. Issiqda yopishqoq, sovuqda mo'rt bo'lib qoladi.

Vulkanizatsiya

Kauchukni **oltingugurt** bilan qizdirish (Ch. Gudyir, 1839). Makromolekulalar orasida –S–S– «ko'priklar» hosil bo'lib, to'rsimon tuzilish yuzaga keladi: mustahkamlik, elastiklik va haroratga chidamlilik ortadi.

Mahsulot	Oltingugurt miqdori	Xossasi
rezina	≈ 1–3 %	elastik, mustahkam
ebonit	≈ 30–40 %	qattiq, elektr izolyatori

Sintetik kauchuklar

Kauchuk	Monomer	Xususiyati, ishlatilishi
butadiyen (SKB)	butadiyen-1,3	birinchi sintetik kauchuk (Lebedev); shina, rezina buyumlar
izopren (SKI)	izopren	tabiiy kauchukka eng yaqin; shina, tibbiy buyumlar
xloropren (neopren)	xloropren	moy, benzin va olovga chidamli; shlang, kabel qobig'i, yelim
butadiyen-stirol (SKS)	butadiyen + stirol	yeyilishga chidamli; avtoshina, poyabzal tagi
butadiyen-nitril (SKN)	butadiyen + akrilonitril	moy-benzinga chidamli; zichlagichlar, shlanglar

7. Ishlatilishi

- **Butadiyen-1,3** – sintetik kauchuklar (SKB, SKS, SKN) va plastmassalar (masalan, ABS-plastik) ishlab chiqarishning asosiy xomashyosi.
- **Izopren** – izopren kauchugi va butil kauchuk (izobutilen bilan) olishda.
- **Xloropren** – moy-benzinga chidamli neopren kauchugi olishda.
- **Rezina buyumlar:** avtomobil shinalari, konveyer tasmalari, shlanglar, poyabzal, tibbiy qo'lqop va buyumlar, sport anjomlari.
- **Ebonit** – elektr izolyatsion materiallar.

Diyenlarni aniqlash: bromli suv va $KMnO_4$ eritmasi rangsizlanadi (alkenlar kabi). 1 mol konyugirlangan diyen 2 mol gacha Br_2 yoki H_2 biriktiradi.

O'zingni tekshir

1. C_5H_8 tarkibli alkadiyenlarning barcha struktura izomerlarini yozing va nomlang.

2. Izoprenning sistematik nomini yozing.

3. Butadiyen-1,3 ga 1 mol Br_2 birikkanda qanday mahsulotlar hosil bo'ladi?

4. Lebedev usuli tenglamasini yozing. 92 g etanoldan necha gramm butadiyen olish mumkin (nazariy)?

5. 5,4 g butadiyen-1,3 bilan necha gramm brom to'liq birikadi?

6. Tabiiy kauchukning tarkibi va tuzilishi qanday?

7. Rezina ebonitdan nima bilan farq qiladi?

Javoblar: 1) 6 ta: pentadiyen-1,2; pentadiyen-1,3; pentadiyen-1,4; pentadiyen-2,3; 2-metilbutadiyen-1,3; 3-metilbutadiyen-1,2; 2) 2-metilbutadiyen-1,3; 3) 3,4-dibrombuten-1 (1,2-birikish) va 1,4-dibrombuten-2 (1,4-birikish); 4) $2C_2H_5OH \rightarrow C_4H_6 + 2H_2O + H_2$; 2 mol etanol → 1 mol, 54 g; 5) 0,1 mol $C_4H_6 \times 2 = 0{,}2$ mol $Br_2 = 32$ g; 6) $(C_5H_8)_n$, sis-1,4-poliizopren; 7) oltingugurt miqdori: rezinada ≈ 1–3 %, ebonitda ≈ 30–40 %.